

Ignite with Robotics & Augmented/Virtual Reality

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Instructions to be followed in all projects:

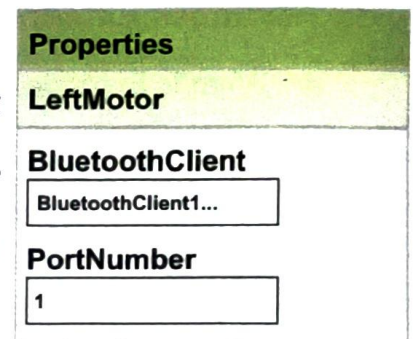
SECTION A

5 Step Guide to Kick start Code 2 Play/The five basic instructions:

In the Designer block:

(Click User Interface under Palette Corner)

1. Drag and drop the **ListPicker** onto the viewer screen, rename it to Connect/Pick Car as required in your project.
2. Drag and drop the **Notifier** onto the viewer screen. (notice it is an invisible component)
(Click Connectivity under Palette column)
3. Drag and drop the **BluetoothClient** onto the screen. (notice it is an invisible component) (Click QtPi Mega Blocks to find all the necessary components you will need for your project)
4. As you add the necessary components for our project from QtPi Mega Blocks, you can **assign the port number** (refer to SECTION B instructions for port connections) and add **Bluetoothclient1** to the respective components used.
5. For all visible and non-visible components, you will have to rename them accordingly in the components and properties columns on the right-hand side of the code2play screen as required in the project.



SECTION B

Instructions for port connections

The QBrik consists of six ports to connect the sensors and actuators. They are P1, P2, P3, P4, P5 & P6. The three ports for motors are M1, M2 & M3 where M3 is a non-programmable port also known as the testing port.

COMPONENT	ACCESSIBILITY OF PORT
IR Sensor	Can access all ports (P1 , P2, P3, P4, P5, P6)
LDR Sensor	Can access only ports (P1, P2, P3, P4)
Ultrasonic Sensor	Can access all ports (P1 , P2, P3, P4, P5, P6)
Temperature Sensor	Can access all ports (P1, P2, P3, P4, P5, P6)
Gas/smoke Sensor	Can access only ports (P1 , P2, P3, P4)
Buzzer	Can access all ports (P1, P2, P3, P4, P5, P6)
RGB LED	In port P2 and P3 all three colors are programmable; In port P1, P5 and P6 Red and Blue are programmable; In port P4 Green and Blue are programmable.
LED	Can access all ports (P1, P2, P3, P4, P5, P6)
LIMIT SWITCH	Can access all ports (P1 , P2, P3, P4, P5, P6)
DC Motor	Can access all motor ports (M1, M2, M3)
Servo Motor	Can access all ports (P1, P2, P3, P4, P5, P6)
Pump Motor	Can access all motor ports (M1, M2, M3)

SECTION C

Instructions for Bluetooth connectivity:

In the Blocks interface

Step 1

On the left-hand side of the screen under **Blocks** column in the **screen 1** section click on connect (ListPicker)

Step 2

Drag and Drop 'when Connect.Beforepicking' and 'when Connect.Afterpicking' Event.

Step 3

- To code the **Beforepicking** event:
 - ◆ Click on **connect(ListPicker)** drag and drop "**set Connect. Elements to**" inside **When Connect. Beforepicking**.
 - ◆ Click on **BluetoothClient1**

- ♦ Drag and attach **BluetoothClient1.AddressesAndNames** to the above.
- To code the **Afterpicking** event:
 - ♦ Use **if-then-else**
- To code the **if** condition
 - ♦ Drag and drop call **BluetoothClient1.Connectaddress** from **BluetoothClient1**.
 - ♦ Click on **Connect(ListPicker)**, drag and drop the **Connect.Selection** code.
- To code the **then** block
 - ♦ Click on **Notifier 1**
 - ♦ Drag and drop call **Notifier1.ShowAlert** notice
 - ♦ Go to **text** drag and drop an empty text string ("")
 - ♦ In the comment type ("**Connected**")
- To code the **else** block
 - ♦ Follow the same steps as **then** block
 - ♦ In the comment type ("**Failed to connect**")

```

when Connect ▾ .BeforePicking
do set Connect ▾ . Elements ▾ to BluetoothClient1 ▾ . AddressesAndNames ▾

```

```

when Connect ▾ .AfterPicking
do if
  call BluetoothClient1 ▾ .Connect
  address Connect ▾ . Selection ▾
  then call Notifier1 ▾ .ShowAlert
  notice " Connected "
  else call Notifier1 ▾ .ShowAlert
  notice " Failed to connect "

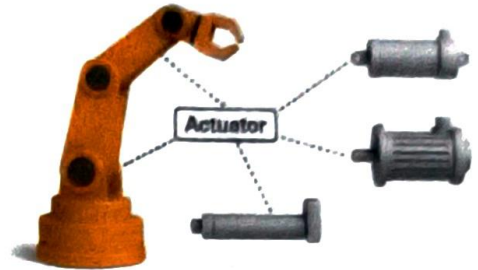
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1

WHAT ARE ROBOTS? UNDERSTANDING THE BASICS

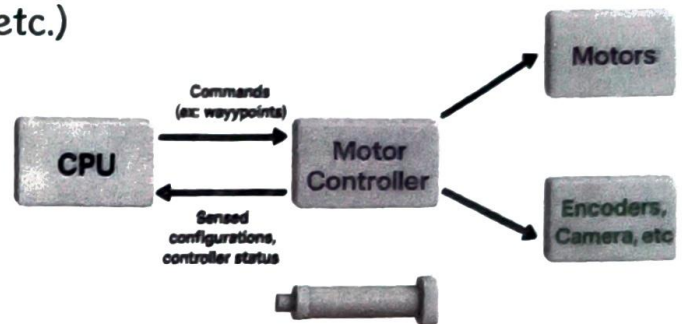
What is a Robot?

A robot is a machine that can sense, think, and act. Robots are used to do tasks that might be dangerous, boring, or hard for humans. Robots work by following a program or using artificial intelligence (AI).



Main Parts of a Robot:

1. **Sensors** – To sense the world (eyes, ears, etc.)
2. **Processor (Controller)** – The robot's brain
3. **Actuators** – Help the robot move
4. **Power Supply** – Provides energy
5. **Body/Structure** – Holds everything



Now to make it even simpler let's read the story

"Meet the Cleaning Team – Robo & Chip!"

Robo is a happy little robot who zooms around cleaning homes. Inside him lives Chip, a clever processor who gives all the commands.

When Robo spots dirt with his sensor, Chip quickly tells him what to do!

Together, they make the perfect team to keep every corner sparkling clean

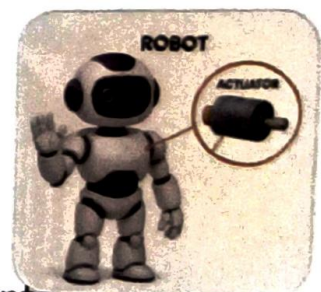


What are Actuators?

Actuators are the parts that make robots move. They turn signals from the processor into physical movement.

Types of Actuators:

- **Electric Motors** – Rotate wheels or gears
- **Servo Motors** – Move parts to a specific angle
- **Stepper Motors** – Move in small precise steps
- **Hydraulic/Pneumatic Actuators** – Use fluid or air for strong movements





EXERCISES

A. Match the Following:

Column A

1. Sensors
- ~~2. Processor~~
3. Actuators

Column B

- A. Help the robot move
- ~~B. Provide strong fluid motion~~
- C. Collect information

Answers

1. C
2. X
3. A

B. Fill in the Blanks:

1. A robot is a machine that can sense, think, and act.
2. The processor is like the robot's brain.
3. Servo motors are used to move parts to a specific angle.



PRACTICALS

Violet's Voice control home automation

It annoyed young Violet to see her friends play outside while she was confined to her bed with a sprained ankle. And it pained her, even more, when she had to limp to get to the door or have the lights and fan switched on. To cheer up her daughter, Violet's mom sat down with her to an episode of Alibaba and the forty thieves. In the middle of the show, Violet commented, "Wouldn't it be nice if I could open the door, and switch on the fan and lights with my voice, just like Alibaba did with the cave?". Then it hit her that she could do it with the right sensors and actuators! So, the next day, Violet grouped her friends to help her in automating her home.

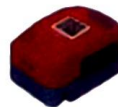
Our home automation system will control the light, fan, and door of the house. It's completely Voice Controlled system e.g. when you say "Fan on" fan turns on.



Materials required



Qbits



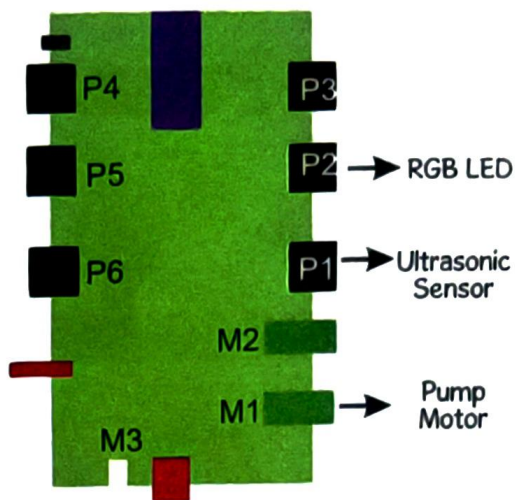
Qbrick



DC Motors



LED



What will you need

Qbits: Qbits are tiny building blocks that are connected to make the skeleton of the robot.

Qbrick: Qbrick is the brain of the robot. It gives command to the Electronics Components and also gives eclectic energy to the components.

DC Motors: A little machine that spins when you give it electricity, helping things move or turn.

LED: LED stands for Light Emitting Diode—it's a tiny light that glows when electricity flows through it, like magic!

BUILDING AND CONNECTIONS

1. First take the Qbits and build a structure of a house and place your Qbrick in the structure you have built.
2. Follow the steps given under Section B (Instructions) to connect the DC Motor, LED and Servo Motor to their respective ports.
For example: (LED to Port P1 and DC Motor to motor port M1)

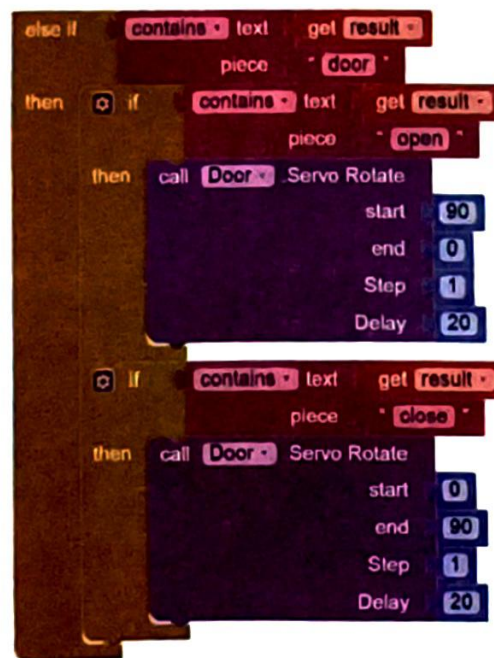
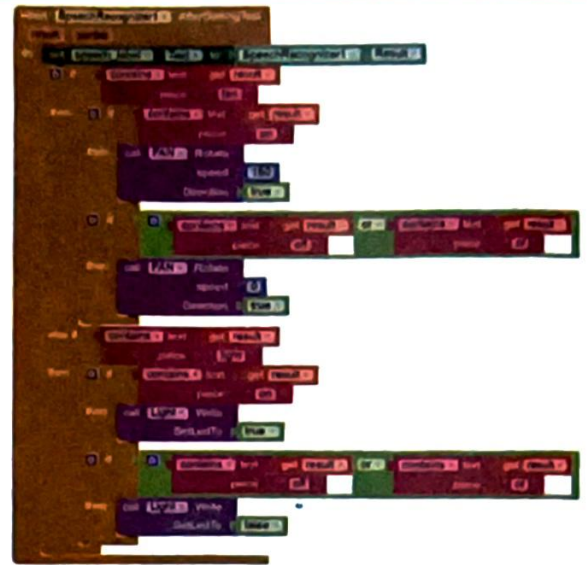
Let's Code

Open Code2Play and follow the steps to build an Home Automation Voice Control

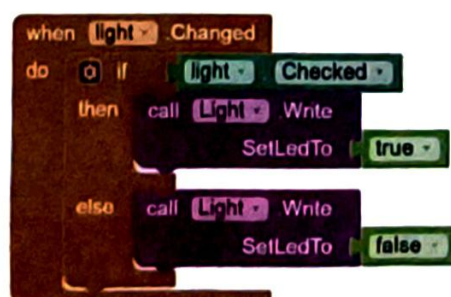
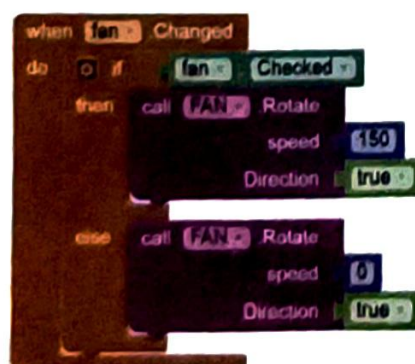
1. *Create a new project.*
2. Add a button – “speak” and pick Speech Recognizer from Media which is vital for recognizing your speech.
3. Add 3 CheckBox's from User Interface and name them as light, fan and door.
4. Complete the Bluetooth Connectivity code. (Refer to **Section C (Instructions)**).
5. When you click on the speak button, a speech recognizer should be called.
6. Click **SpeechRecognizer1.AfterGetting Text** event.



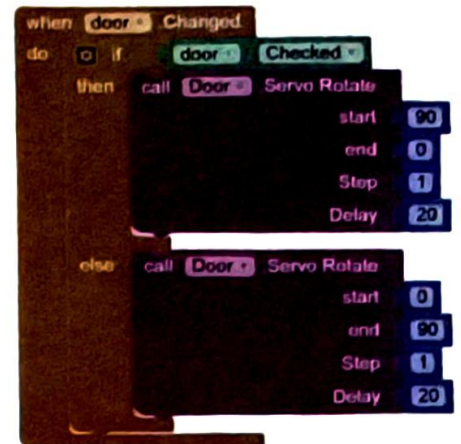
7. Add a Label to display the voice command.
Drag and drop **set speech_label.text** to inside the **SpeechRecognizer1.AfterGetting Text** event and attach **SpeechRecognizer1.Result** to it.
8. Compare and check if the voice command is "switch ON the fan" Or "switch OFF the fan".
9. Assign the speed and direction of the fan accordingly. Similarly, check if the voice command is to "switch ON the light" Or "switch OFF the light" to complete the program
10. Similarly, complete the logic for the door, if the voice command is to "open the door" or "close the door" assign the values for start, end step and delay of the door accordingly.



11. Next, click on the fan checkbox and select when fan.Changed event and add it to the if-then-else block.
12. Attach fan.Checked from fan Checkbox to the if statement.
13. Next, drag and drop call **FAN.Rotate** and assign the speed and direction of the fan accordingly.



14. Similarly, write the logic for Light and Door.



PLUG AND PLAY

Turn on your Qbrick, open the Home Automation Voice Control app in your android device and connect your Qbrick to it and enjoy playing with your Home Automation with Voice Control!

What do you understand from this project?

Similar applications

1. **Home Control App:** Students can turn lights or fans on and off by giving voice commands.
2. **Elderly Help App:** Old or disabled people can use it to control appliances without moving.

Result of this project

1. **Voice Command Result:** Lights or fans turn ON and OFF when students give voice commands.
2. **Convenience Result:** Students can control appliances without touching switches, making it simple and fun.

2

How Robots Move: Sensors and Processors Working Together

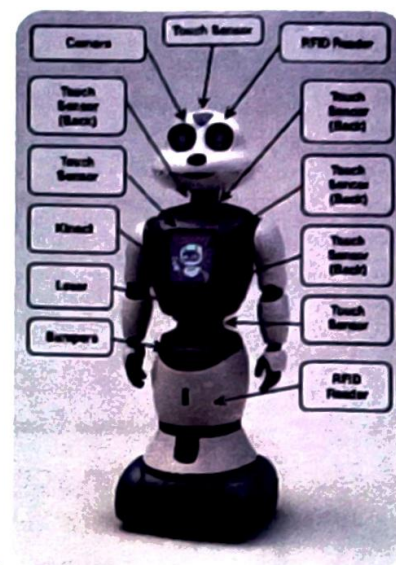
How Do Sensors and Processors Work Together?

Robots follow a simple cycle to perform tasks:

Sense → Think → Act

Step-by-Step:

1. Sensor collects data – like distance, touch, or light.
2. Data goes to the processor – the robot's brain.
3. Processor uses a program to decide what to do.
4. Sends signal to actuator (like a motor).
5. Actuator performs the action – moving, stopping, turning, etc.



Real-Life Example:

An ultrasonic sensor detects a wall. The processor reads this information and decides to stop or turn. It sends a signal to the motor, and the robot moves in a different direction.

Types of Sensors:

Ultrasonic Sensor – Measures distance

Touch Sensor – Detects contact

Light Sensor – Detects brightness

IR Sensor – Detects infrared light (used to follow lines or avoid edges)

Types of Motors and Their Specifications:

Motor Type	Voltage Range	Special Use	Description
DC Motor	3V–12V	Wheels, general motion for continuous rotation	Basic motor
Servo Motor	4.8V–6V	Arms, joints	Moves to specific angles
Stepper Motor	5V–12V	Precise positioning great for accuracy	Moves in steps,
Gear Motor	6V–24V	High torque applications	Contains gears for powerful movement
Brushless Motor	7.4V–14.8V	Drones, high-speed devices	High speed, no brushes, longer lifespan



ACTIVITY

Create a Flowchart that shows:

Sensor → Processor → Actuator → Action

Motor Matching Game: Match motors with their uses:

- Servo Motor → _____
- Stepper Motor → _____
- Gear Motor → _____



EXERCISES

A. Match the Following:

Column A

1. Ultrasonic Sensor
2. Processor
3. Touch Sensor

Column B

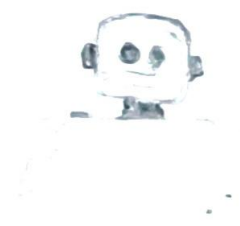
- A. Turns signal into movement
- B. Measures distance
- C. Thinks and controls the robot

Answers

1. B
2. C
3. A

B. Fill in the Blanks:

1. Sensors help the robot sense the world.
2. A processor makes decisions using the program.
3. Actuators get instructions from the robot.



C. Multiple Choice Questions (MCQs):

1. What happens first in the Sense-Think-Act cycle?

☐ A. Think ☐ B. Act ☒ C. Sense ☐ D. Move

2. Which sensor is used to avoid obstacles?

☐ A. Light sensor ☒ B. Ultrasonic sensor ☐ C. Sound sensor ☐ D. IR sensor

3. What does the processor use to make decisions?

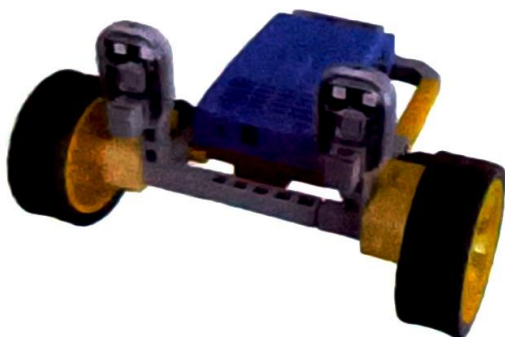
☐ A. Heat ☐ B. Light ☒ C. Program ☐ D. Touch



PRACTICALS

Rishi's RC Car with an indicator

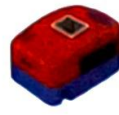
Rishi was very fond of playing with his remote-controlled car. What he wasn't fond of was his brother, Roshan, crashing his own car into his at every corner of the corridor. This would lead to the brothers arguing over who got to race down the corridors with their car first while the other sat down for his turn. Feeling guilty for taking up his brother's playtime, Rishi sat down to come up with something to indicate Rohan of his car's presence so that they could play with their car at the same time and avoid a collision.



Materials required



Qbits



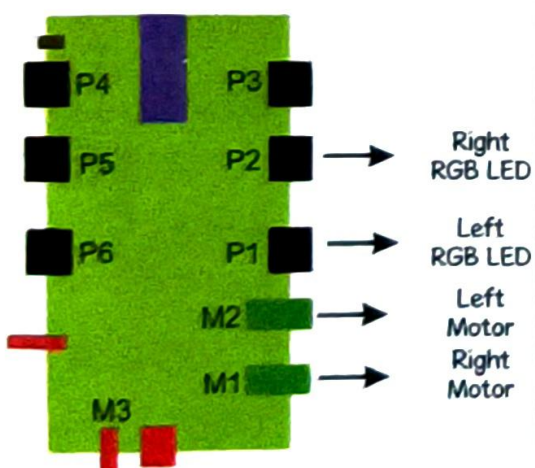
Qbrick



DC Motors



RGB LED - 2



What will you need

Qbits: Qbits are tiny building blocks that are connected to make the skeleton of the robot.

Qbrick: Qbrick is the brain of the robot. It gives command to the Electronics Components and also gives eclectic energy to the components.

DC Motors: A little machine that spins when you give it electricity, helping things move or turn.

RGB LED: RGB stands for Red, Green, and Blue—the three magic colors that mix together to make millions of other colors on screens and lights!

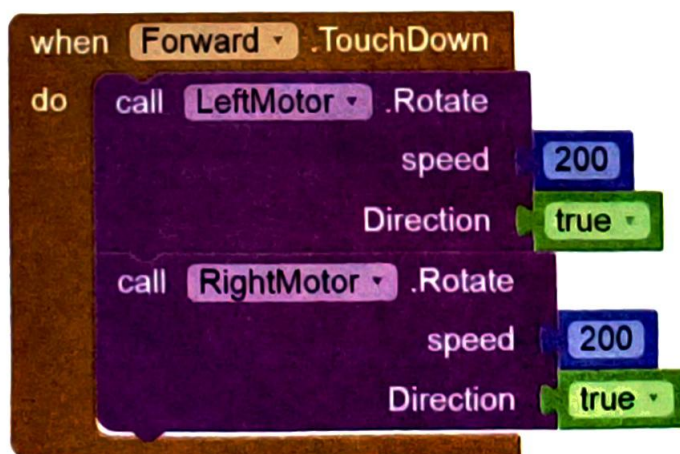
BUILDING AND CONNECTIONS

1. First, take the Qbits and build a structure of RC Car and place your Qbrick in the structure you have built.
2. Follow the steps given under Section B (instructions) to connect the two DC motors and two RGB LED to their respective ports.
For example: (RGB1 to Port P1 and RGB2 to port P2; left DC Motor to motor port M1 and right DC Motor to motor port M2)

Let's Code

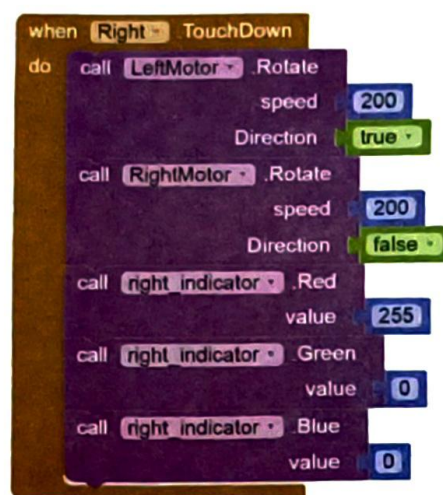
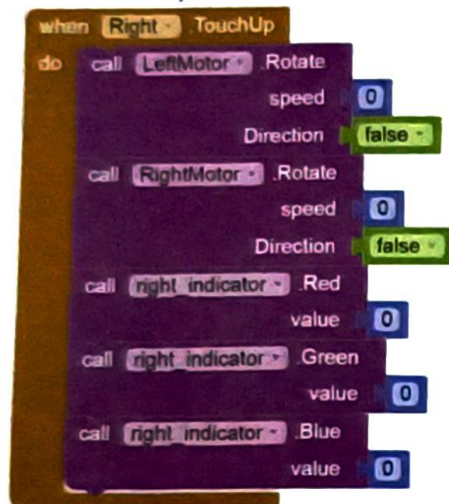
Open Code2 Play and follow the steps to build an RC Car with an indicator app.

1. Add the required components for the project. (Refer to **Section A (Instructions)**).
2. In the Designer interface add four buttons from the User Interface palette.
3. Complete the **Bluetooth Connectivity** code. (Refer to **Section C (Instructions)**)
4. Now, click the Forward button and choose the **when Forward. TouchUp** and **when Forward. TouchDown** events.
5. Now, drag and drop the **call LeftMotor. Rotate** and **call RightMotor. Rotate** blocks and assign the speed and direction for **TouchDown** and **TouchUp** event as shown in the figure.
6. Next, in **Reverse, Left** and **Right** buttons, choose the **TouchUp** and **TouchDown** events.
7. Now program in such a way that:
 - When the Forward button is touched, RC Car should move forward and both RGB LEDs should be turned OFF.



- When the Reverse button is touched, Car should move backward and both RGB LED should be turned OFF.
- When the Right button is touched, Car should move right and the right side RGB LED should be turned ON.
- When the Left button is touched, Car should move left and the left side RGB LED should be turned ON.

Now let's try to do this!



What do you understand from this project?

Similar applications

1. **Traffic Learning App:** The RC car can teach students about road safety and traffic rules using indicators.
2. **Mini Demo App:** The car can be used in school to show how real vehicles signal turns.

Result of this project

1. **Movement Result:** The RC car moves forward, backward, left, and right when controlled.
2. **Indicator Result:** The car's lights blink left or right to show the direction, just like real cars.

3

HOW DOES EXTENDED REALITY WORK?

What is Extended Reality (XR)?

Extended Reality (XR) is a term that includes **Virtual Reality (VR)**, **Augmented Reality (AR)**, and **Mixed Reality (MR)**.

Understanding Extended Reality (XR)

XR includes:

Virtual Reality (VR): Virtual Reality is like stepping into a whole new world that's not real! You wear a special headset that covers your eyes, and suddenly, you're in a completely different place. It could be a video game world, outer space, or even a fantasy world!

For Example, Think of a video game where you wear a headset and feel like you're in the game, swinging a sword or flying a spaceship. That's VR!

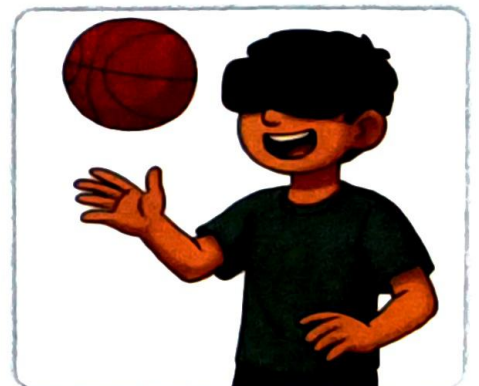


Augmented Reality (AR): Augmented Reality adds things to the real world around you. Imagine looking at your classroom through your phone, and suddenly, an animated character appears on your desk! It's still the real world, but AR adds digital things to it.

For Example, Apps like Pokémon Go use AR. You can see Pokémon characters in the real world through your phone screen, like they are right in front of you.

Mixed Reality (MR): Mixed Reality is a combination of VR and AR. It's like the best of both worlds! It doesn't just put digital things into the real world; it lets those things interact with the real world.

For Example, imagine you're wearing special glasses and see a virtual basketball in front of you. When you move your hands, the ball reacts—bouncing or rolling just like a real one.



Difference between Augmented Reality (AR), Virtual Reality (VR) and Mixed Reality (MR)

Feature	Augmented Reality (AR)	Virtual Reality (VR)	Mixed Reality (MR)
Immersion Level	Low	High	Medium to High
Real-World Interaction	Yes	No	Yes
Required Hardware	Smartphone, AR Glasses	VR Headset, Controllers	MR Headset, Sensors
Example Device	Google Lens, Snapchat	Oculus Quest, HTC Vive	Microsoft HoloLens, Magic Leap
Example Apps	IKEA Place, Google Maps AR	Beat Saber, VRChat	HoloAnatomy, Spatial

Hardware Components of XR

1. Head-Mounted Displays (HMDs)

- **Function:** These are wearable devices that present digital content directly to the user's eyes.
- **Examples:**
 - **VR Headsets:** Oculus Rift, HTC Vive.
 - **AR Glasses:** Microsoft HoloLens, Google Glass.
- **Features:** High-resolution displays, wide field of view, and motion tracking sensors.

2. Tracking Systems

- **Function:** Monitor the user's movements and translate them into the digital environment.
- **Types:**
 - **Inside-out tracking:** Sensors on the device track movement.
 - **Outside-in tracking:** External sensors track the device.
- **Importance:** Ensures accurate alignment of virtual content with the real world.

3. Controllers and Input Devices

- **Function:** Allow users to interact with virtual environments.
- **Examples:** Oculus Touch controllers, HTC Vive wands, haptic gloves.
- **Features:** Buttons, touchpads, motion sensors for intuitive control.

4. Haptic Feedback Devices

- **Function:** Provide tactile sensations to simulate touch or pressure.
- **Examples:** Teslasuit, HaptX Glove.
- **Use Cases:** Enhance realism in training simulations, gaming, and remote operations.

5. Cameras and Sensors

- **Function:** Capture real-world data for spatial awareness and interaction.
- **Types:** Depth cameras, infrared sensors, motion detectors.
- **Role:** Essential for AR and MR experiences to overlay digital content accurately.

Software Components of XR

1. Development Platforms and Engines

- **Function:** Tools for creating and rendering XR content.
- **Examples:** Unity3D, Unreal Engine, ARKit.
- **Capabilities:** Support for 3D modelling, animation, and interactive experiences.

2. Software Development Kits (SDKs)

- **Function:** Provide libraries and tools for developing applications on specific hardware.
- **Examples:** Oculus SDK, Google ARCore, Microsoft Mixed Reality Toolkit.
- **Purpose:** Simplify the development process and ensure compatibility.

3. User Interface and Interaction Libraries

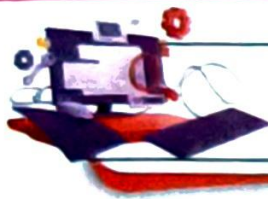
- **Function:** Help design intuitive user interfaces and interactions in XR.
- **Examples:** Mixed Reality Toolkit (MRTK), Virtual Reality Toolkit (VRTK).
- **Benefits:** Facilitate natural interactions like gestures and voice commands.

4. Spatial Computing and Mapping

- **Function:** Allow devices to understand and map the physical environment.
- **Technologies:** Simultaneous Localisation and Mapping (SLAM).
- **Importance:** Crucial for accurate placement of virtual objects in AR and MR.

5. Content Creation and Management Systems

- **Function:** Tools for creating, storing, and managing XR content.
- **Examples:** 3D modelling software like Blender, content distribution platforms like SteamVR.
- **Role:** Enable developers to produce and share immersive experiences.



EXERCISES

A. Fill in the Blanks

1. A game like Pokémon GO uses _____ Reality.
2. XR is used for learning, games, and even _____.
3. XR includes VR, AR, and _____ Reality.

B. Choose the correct answer

1. AR works without any devices. ☐ a) True ☐ b) False
2. VR stands for Very Real. ☐ a) True ☐ b) False

C. Short Questions and Answers

1. Is XR only for entertainment?

2. What is one fun use of VR?

3. What are the three types of XR?



PRACTICALS

A. Visit a theme park in VR

Click the link below to take a thrilling Virtual Reality (VR) ride through Jurassic World. Let us experience the excitement of coming face-to-face with dinosaurs.

Visit [Jurassic World](#) in your AI Skill lab.



Once we open the clip on the top left there is a navigation point (highlighted with red circle) by pressing the arrow (left, right, top and bottom) look all around.

B. Visit a Wonder of the World in VR

Click the link below to take a thrilling Virtual Reality (VR) visit Rio de Janeiro. Let us experience the excitement of coming face-to-face with one of the seven wonders of the world.

Visit [Rio de Janeiro](#) in your AI Skill lab.



4

INTRO TO 3D BLOCK-BASED CODING

Introduction to Tinkercad

Tinkercad is a user-friendly, web-based platform that allows users to create 3D designs. It's widely used in schools to help students learn the basics of 3D modelling.

Creating a 3D Model in Tinkercad

Open Tinkercad:

Log in to Tinkercad: Click on the "Log in" button on the top right.

Sign in with Google:

- Choose "Personal Account".
- Select "Sign in with Google" and pick your

Gmail account.

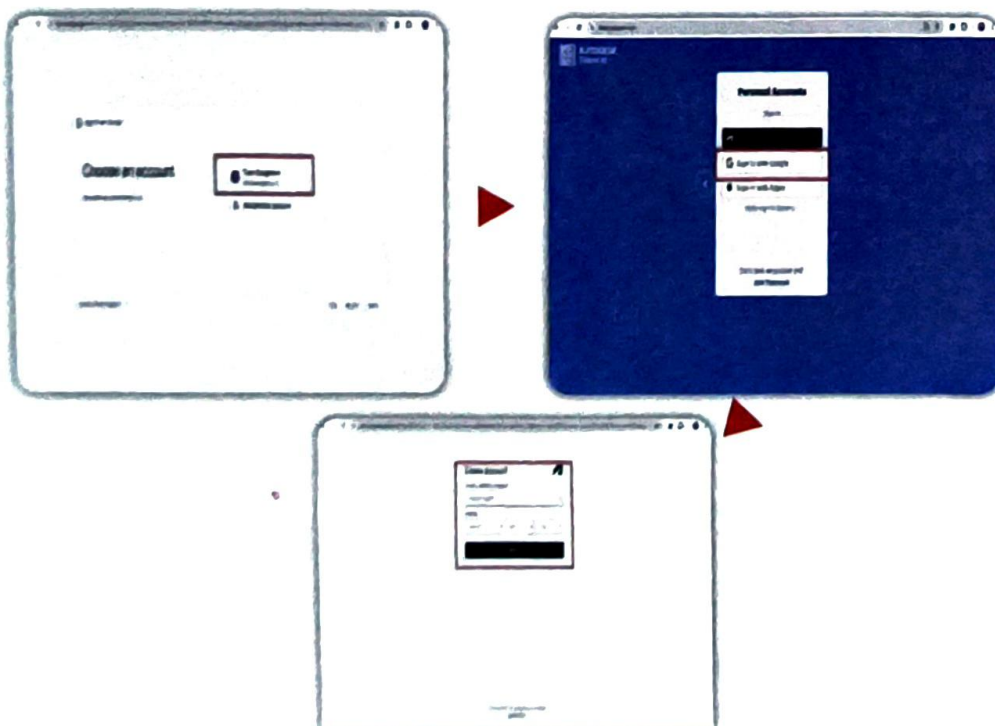
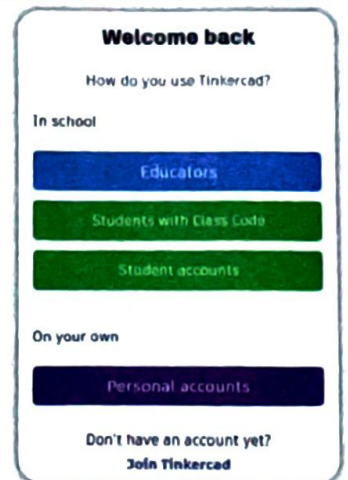
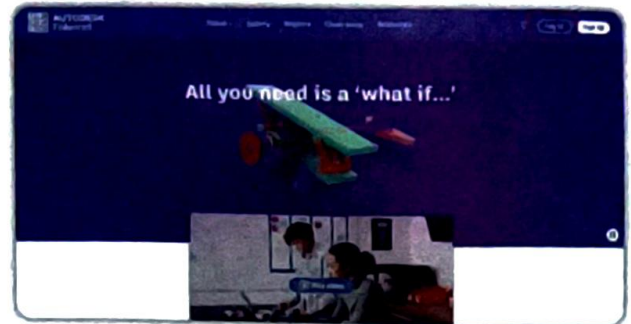
- Choose the Gmail account you created or an existing one.

Enter Personal Details:

Provide birth date and country of origin when prompted.

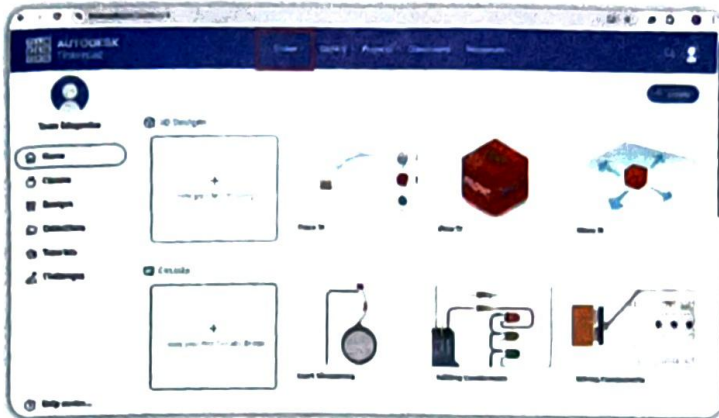
Enter Personal Details:

- Provide birth date and country of origin when prompted.

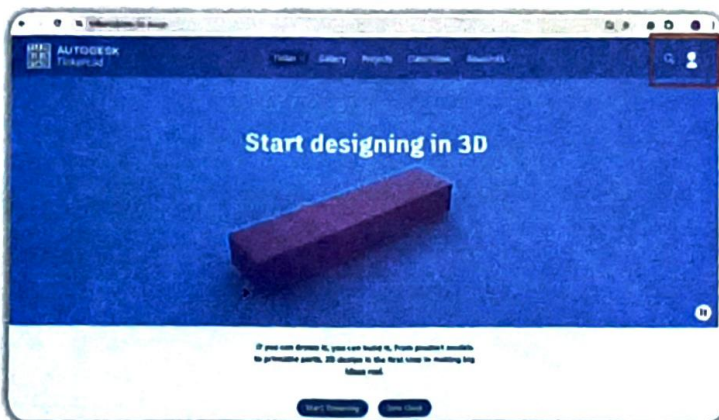


Creating a pond activity by using Tinkercad

- Once logged in, we can access the dashboard. Then click on the "Tinker" section.
- Select "3D Design" to begin your project.



- On your far right, there is a search bar tool. Click on it and type "Pond" to access ready-made pond projects.

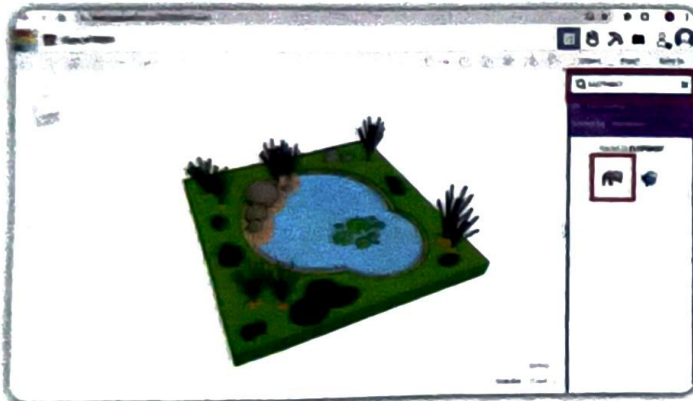


- Search for a pre-made pond model. Please choose the project displayed in the image above.
- Click on the search bar, you will get options like the one mentioned in the image below select the Creatures & Characters option.

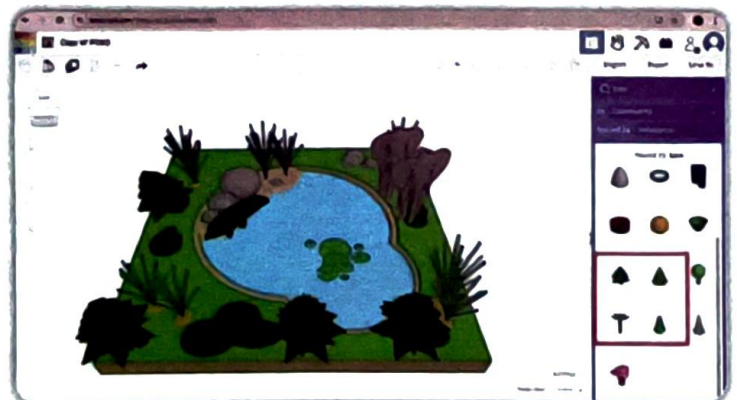


Add Creatures to the Pond.

- Use the search bar to find "Creatures & Characters."
- Type "Elephant" in the highlighted Search Shapes bar and select the highlighted elephant figure. You can add as many creatures and trees as you want.
- Hold the elephant and drop it over your pond.



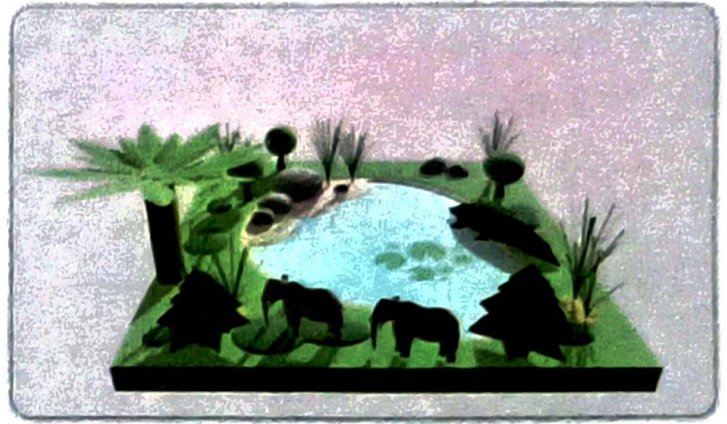
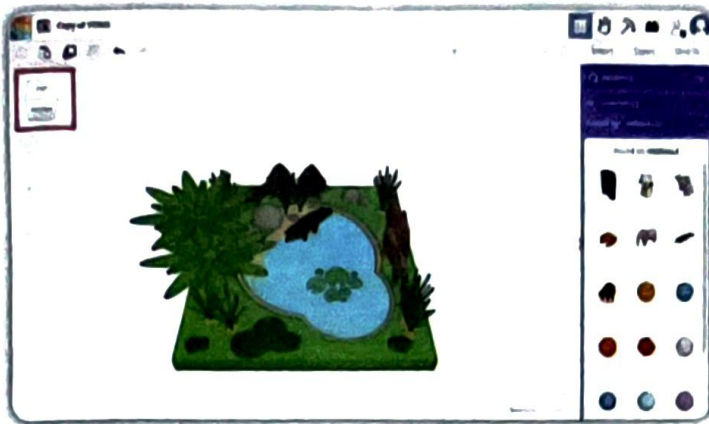
- Using the square key, change the size and shape of the objects placed as per your convenience.
- Use the rotating control point to change the direction of the object.



- Add some trees as shown in the picture and Crocodiles, and place them as illustrated in the picture below. You can add as many animals and tree you want to add in your 3D Model.



- Use the circled controller to see your model from different angles.



- Once you have finished building the project, export it as an OBJ file. This will allow you to share and use your work easily in other applications.

Export the Project.

- Click on the "Export" button (top-right corner).
- In the export options, select "OBJ" format.
- Tinkercad will generate the OBJ file and start downloading it to your computer.



VR Development Using Sketchfab

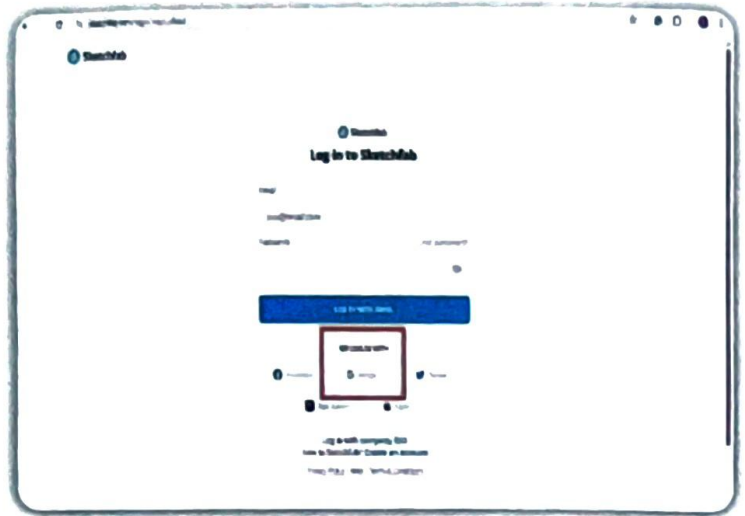
Introduction to VR/AR Settings

Sketchfab provides built-in VR (Virtual Reality) and AR (Augmented Reality) tools that allow you to fine-tune how your 3D models appear when viewed through a VR headset or AR-compatible device.

- These settings help with:
 - Correct scale – making sure your model appears life-sized.
 - Proper alignment – placing your model at the correct eye level or ground level.
 - Immersive experience – ensuring it feels realistic in VR/AR.
- Only the creator of the model can access and adjust these settings. These changes do not alter your original file, but they do affect how others view it in 3D/VR/AR.

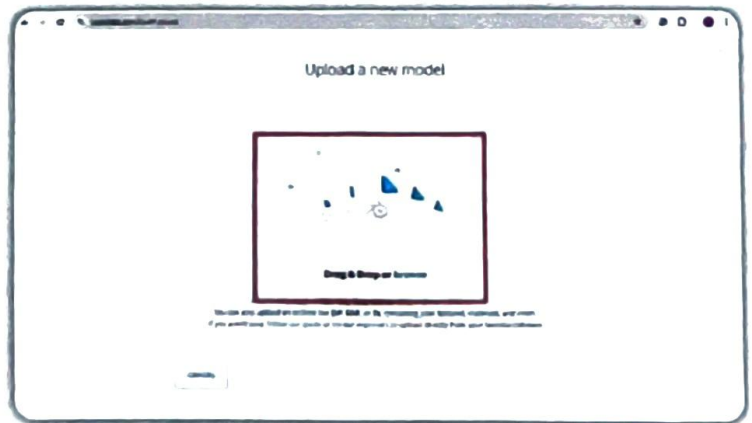
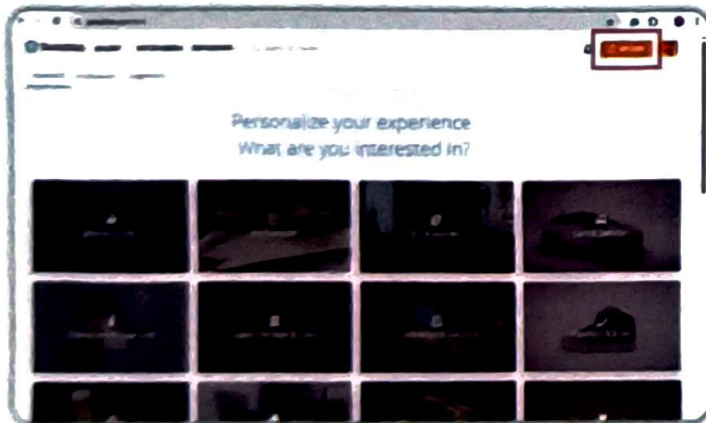
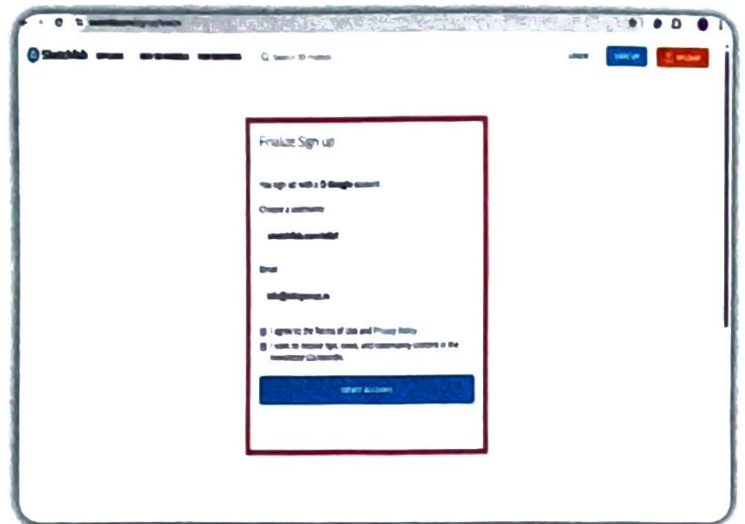
Create and Verify Your Gmail to Set Up Your Sketchfab Account

- Open your browser and go to Sketchfab.
- Sign in to your Sketchfab account using your Gmail or preferred method.
- If you don't have an account, click on Sign Up and follow the process to create one.



Uploading Your 3D Model to Sketchfab

- Once you're logged into your Sketchfab account, click the "Upload" button at the top right of the screen.
- Choose "Upload a 3D model".
- Browse your computer to find and select your.OBJ, .GLTF, or.FBX 3D file that you exported from Tinkercad or any 3D tool.
- Wait for the model to finish uploading.

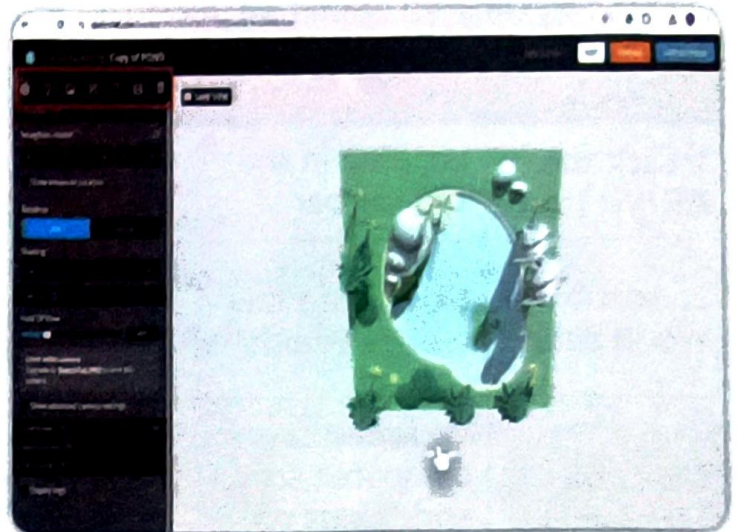


- Once the upload is complete, your model will be processed and viewable in 3D format on Sketchfab.

Edit 3D Settings

- After your model is uploaded and published, click on the "Edit 3D Settings" button.
- This button appears below your model viewer on the Sketchfab page.
- This will open a new interface where you can:

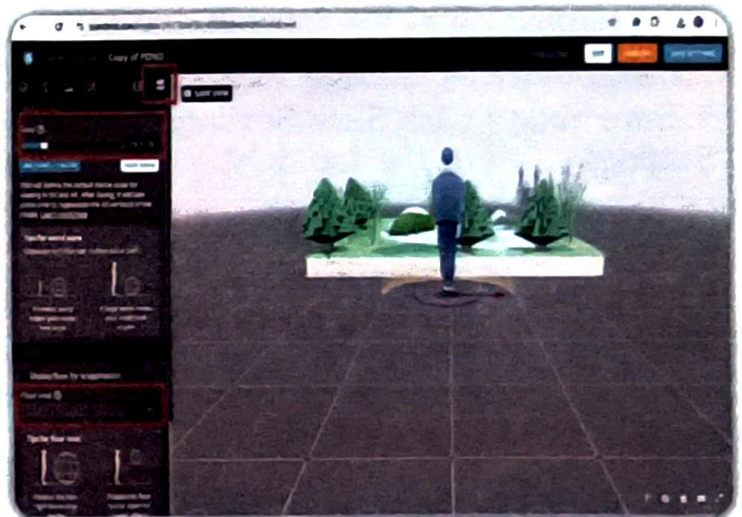
- Adjust lighting
- Change the background
- Edit materials
- And most importantly, configure VR/AR settings for your model.
- Take your time to explore the options and improve how your model looks and behaves in 3D and VR views.



Enable Advanced Rotation & Adjust Floor Level.

Enable the Advanced Rotation feature (toggle switch available near rotation controls).

- Use the blue floor line to align your model with the ground level.
- This helps the model appear naturally placed when viewed in VR.
- Now your model should be properly straightened and grounded, ready for an immersive VR experience!



What You're Seeing

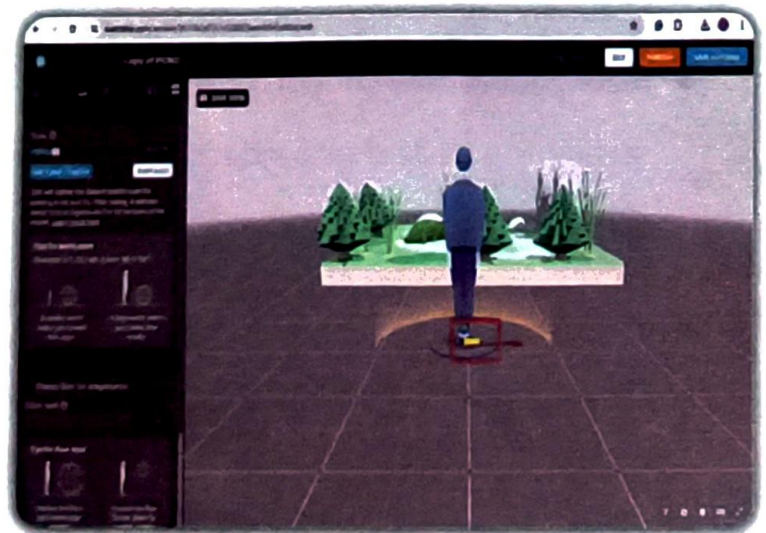
- A 3D model on a grid surface. A blue human figure standing on the model — this represents the average viewer's height in VR (1.75m).
- Two key settings are highlighted:
- World Scale – Controls the overall size of the model in VR/AR.
- Floor Level – Adjusts where the virtual ground (floor) is for teleportation or standing.

How to Set Up a 3D Model for AR/VR in Sketchfab

Steps	Action	Explanation
1. Click the AR/VR Icon	Located in the top toolbar.	Opens the settings that help you prepare your scene for immersive environments.
2. Adjust the World Scale	Move the slider under "Scale".	Determines how big or small the model will appear in VR/AR. A larger scale makes the model feel smaller, and vice versa.
3. Set the Floor Level	You see the computer screen and things on the desktop.	You see a whole new world that looks like a video game or another place.
4. Use the Avatar Preview	The blue figure helps you visually test the scale and floor level.	Check if the character is standing at the correct height and appears realistically sized in the environment.
5. Save Your Settings	Click Save Settings at the top right.	This ensures your AR/VR setup is saved with the model.
6. (Optional) Publish the Model	Use the Publish button if you're ready to share it online.	Once published, the model can be viewed in VR headsets or AR-compatible devices.

Preview Your Model in VR/AR Mode

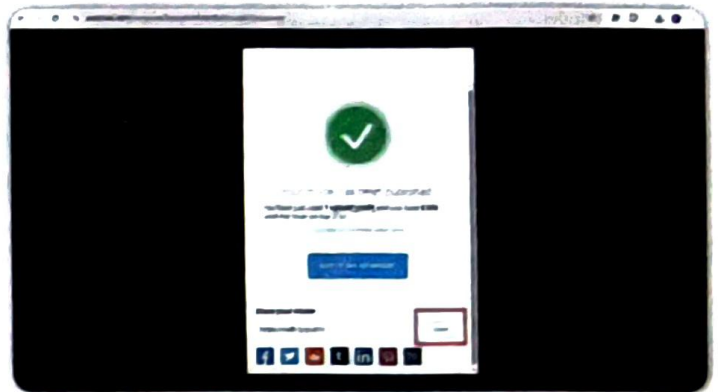
- In the 3D Settings panel, click on the "VR/AR" tab.
- This section allows you to preview how your model will look inside a VR device.
- Look for the VR viewer window to get a real-time idea of the model's size and placement.



Save and Publish Your VR Model

- Once your model is fully adjusted (scale, floor level, viewer position), click on the "Save View" option.
- This saves the current camera angle and position, so when someone opens your model, they see it exactly how you intended.

- Below the preview area, look for a small VR headset icon.
- Click on it to experience a preview of how the model will look inside a VR headset.
- This helps you check the user's perspective one final time.
- After confirming everything looks good:
- Click on "Publish" to finalise and make your 3D model live.
- Now your model is ready to be used in Median to build your VR app.
- You're now ready for the next step copying the model link and creating your app!



Introduction to Median

The Median app is a platform that enables users to convert their existing websites into fully functional native mobile applications for iOS and Android.

Creating a Median Account to Build Your VR App

Open the Median Website

- Go to [Median](#) in your AI skill lab.

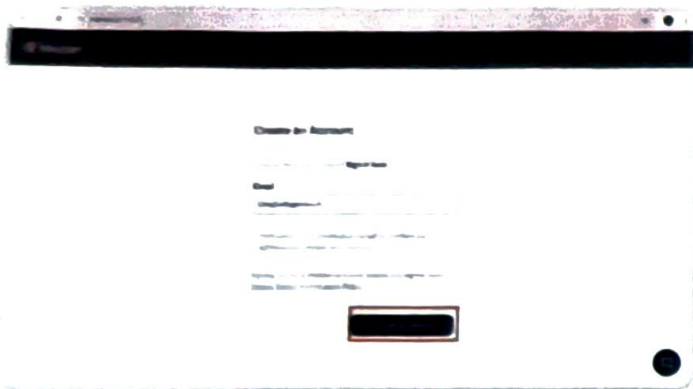
Sign In / Sign Up

- Click on the "Sign In" button at the top-right corner.
- If you already have an account, log in with your credentials.
- If you're new to Median, click on "Sign up" or use Google sign-in to create your account quickly.

Verify Your Gmail

- Google may ask you to enter your password or verify your identity.
- Complete the verification to successfully sign in to Median.





- Enter Your Personal Details and Activate Your Account
- After signing in, you'll be asked to fill in your personal details such as:
 - o Full Name
 - o Organisation Name (if applicable)
 - o Role or Position (e.g, Student, Educator)
 - o Phone Number (optional)

Refer to the image provided for guidance on where to enter each detail.

- Once done, click on "Activate" or "Continue" to complete your account setup.

Create a New App

Once your Mediant account is set up:

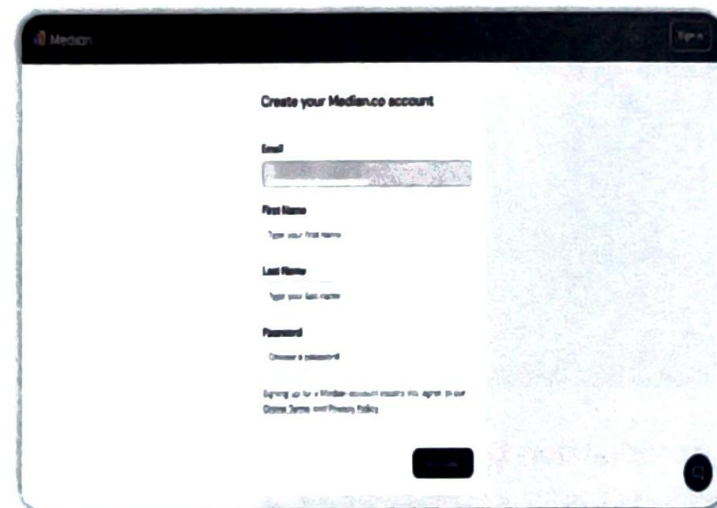
1. You'll land on the dashboard (like the one shown in the image).
2. Click on the "Create a new app" button highlighted in red.

This will take you to the setup screen where you'll enter your app link, name, and organisation details.

Link Your 3D Model & Customise the App

Now it's time to connect your 3D model and personalise your app:

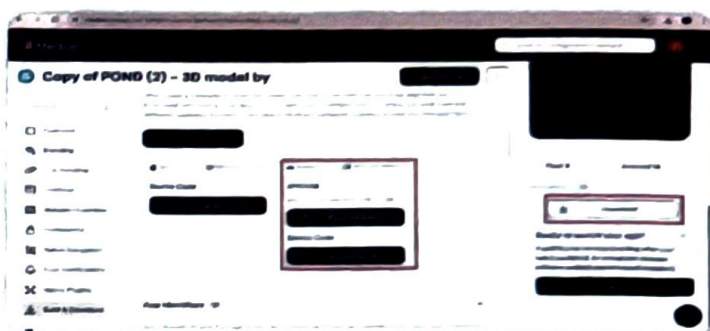
1. Copy the link of your 3D model from Sketchfab.



2. Paste the link in the Website URL field as shown in the image.
3. You can edit the App Name to make it easy to recognise (optional).
4. Under Organisation, your team name (e.g., "Eugenius's Team") will be selected by default.
5. Click on "Start Building my App!" (Highlighted in black).

Phone Simulator Settings

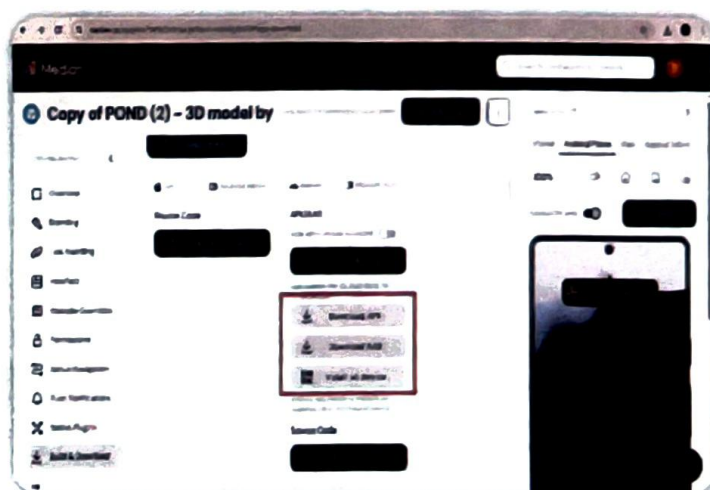
On the right-hand side, you can preview how the app looks. Change the simulator to match your device — choose android phone or iPhone depending on what you're using. Choose android and proceed further.



Download & Build Your App Binaries

Now that your model link is added and your app is set up, it's time to build the app so it can be downloaded:

1. Scroll down the page after clicking "Start Building my App".
2. Look for the "Build & Download" section in the left-hand menu.
3. Click on it, then select the option to "Build Binaries".
4. The platform will now start generating your app package. This might take a few minutes.



Your screen should look something like the image shown, with options for building and downloading the app.

Install the App on Your Phone

- First, install a QR Scanner app from the Google Play Store (for Android) or App Store (for iPhone), if your phone doesn't already have a built-in QR code scanner.
 - You can search for "QR & Barcode Scanner" and install any trusted app.

Scan the QR Code to Download Your App

1. Open the QR scanner app you installed earlier on your Android smartphone.
2. Once your Median app is built, click on "Install on Device."

3. A QR code will appear on your screen.
4. Point your camera at the QR code displayed on the Median platform.
5. The scanner will automatically detect the link and prompt you to download the app.
6. Follow the instructions to install the app on your phone.
7. This will download and install your custom 3D model app directly onto your Android or iPhone device.

You're now ready to open your app and explore the 3D model using your phone or VR headset!

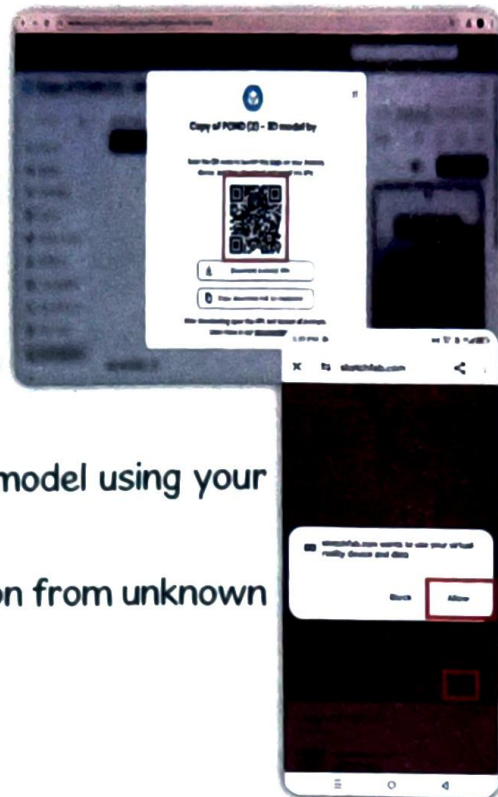
Note for Android users: You might need to allow installation from unknown sources in your settings.

View Your 3D Model in Virtual Reality

Your final result should look like the image shown here.

1. Open the installed app on your Android phone.
2. Place your phone inside a VR headset (like Google Cardboard or any compatible VR device).
3. Put on the VR headset and look around to explore your 3D model in immersive virtual reality!

Congratulations! You've successfully built and experienced your own VR app using Median and Sketchfab.



A. Fill in the Blanks

1. XR applications in _____ allow users to try on clothes and accessories virtually, enhancing the online shopping experience.
2. XR provides an _____ learning experience, engaging multiple senses.

B. Choose the correct answer

1. VR can be used to simulate dangerous situations for training purposes.

☐ a) True ☐ b) False

2. Students can use XR to explore historical events and locations.

☐ a) True ☐ b) False

C. Short Question and Answers

1. Write any 5 applications of XR?

2. What is a virtual field trip?



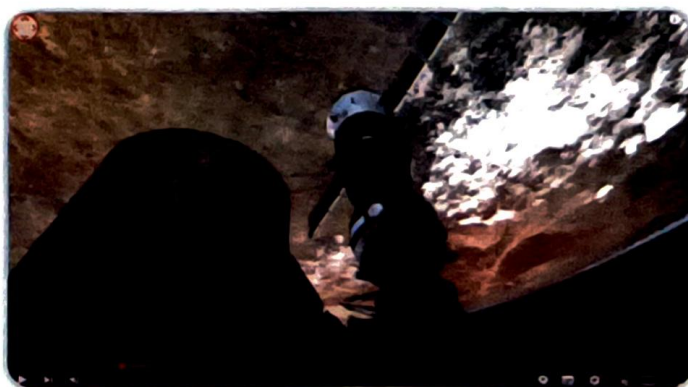
PRACTICALS

A. Try out the following in the AI Skill lab.

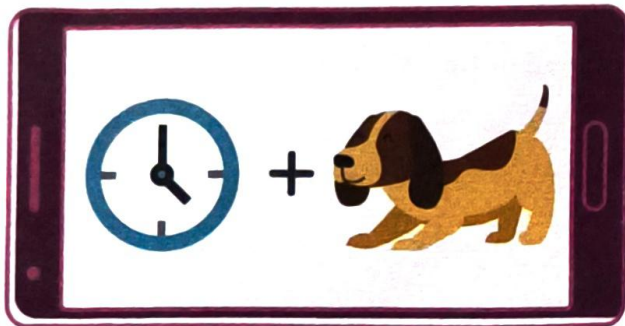
Click on the link below to experience an exciting VR spacewalk adventure and see what it's like to float in space just like an astronaut!"

Visit [spacewalk adventure](#) in your AI Skill lab.

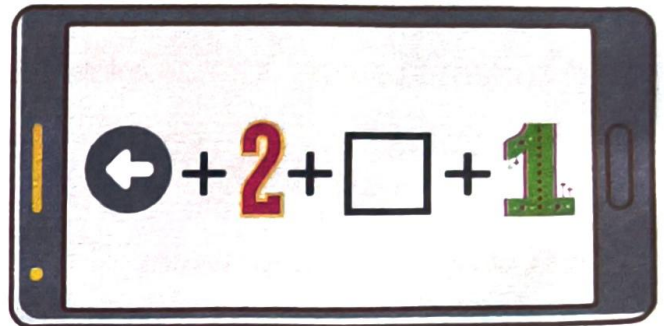
Once we open the clip on the top left, there is a navigation point (highlighted with a red circle). By pressing the arrow (left, right, top and bottom), look all around.



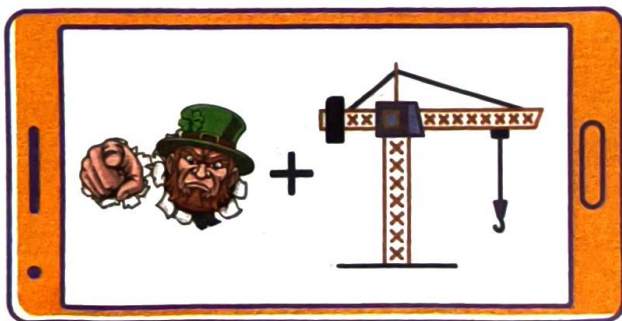
GUESS WHAT?



Guess the word



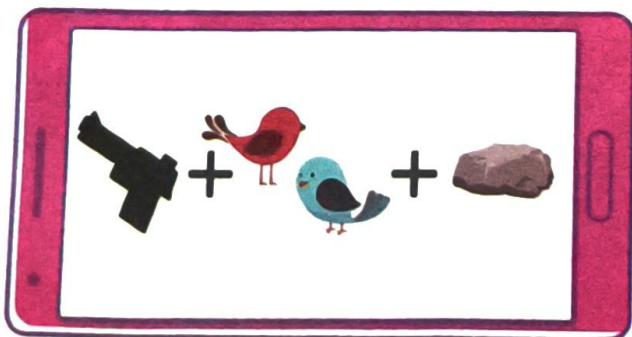
Guess the phrase



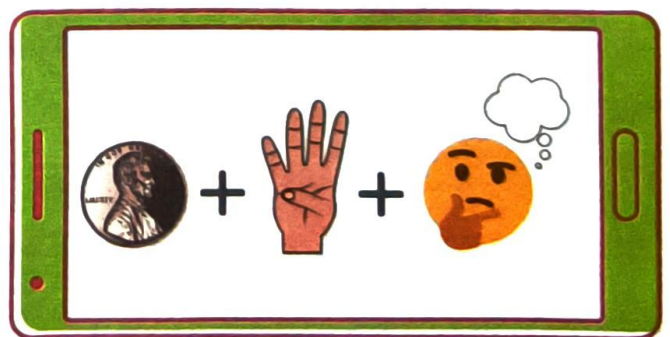
Guess the word



Guess the word



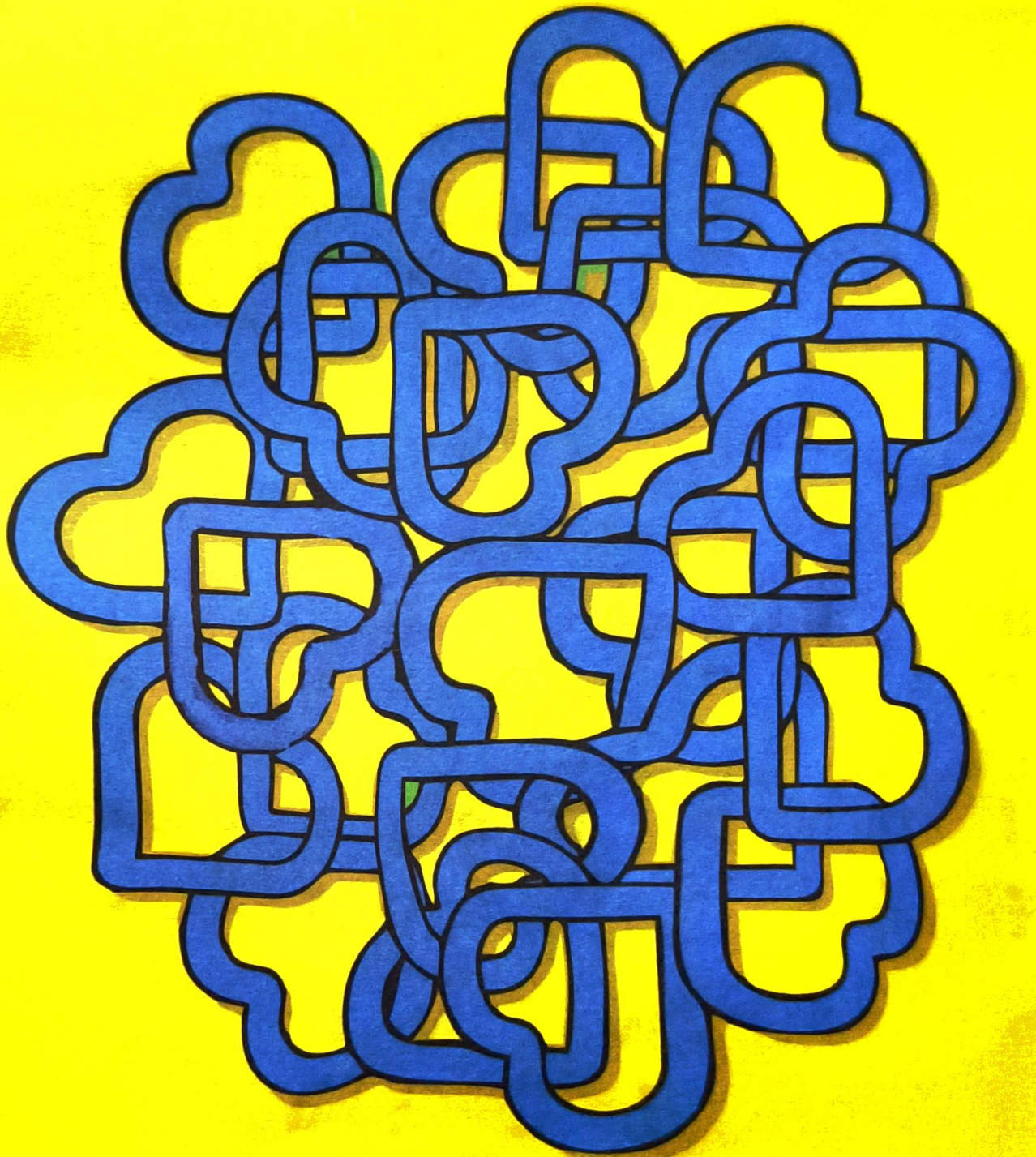
Guess the idiom



Guess the idiom

LINKED LOOPS

Find the 4 pairs of interlinked heart loops from the 2 dozen tangled hearts.



BRAILLE

Braille is a tactile system of reading and writing for use by the blind or visually impaired. Braille is named after its creator Louis Braille (1809 – 1852). In 1824, at the age of 15, Braille presented his ideas for a system of tactile code that could allow blind people to read and write quickly and efficiently.

It consists of raised dots arranged in "cells." A cell is made up of six dots that fit under the fingertips, arranged in two columns of three dots each. Each cell represents a letter, a word, a combination of letters, a numeral or a punctuation mark.

The Braille Cell

1 ● ● 4

2 ● ● 5

3 ● ● 6

The Braille Alphabets

a	b	c	d	e	f	g	h	i	j	k
⠁	⠃	⠉	⠙	⠑	⠋	⠗	⠓	⠊	⠚	⠅

l	m	n	o	p	q	r	s	t	u	v
⠌	⠍	⠎	⠕	⠏	⠑	⠗	⠓	⠔	⠥	⠦

w	x	y	z
⠪	⠭	⠽	⠵

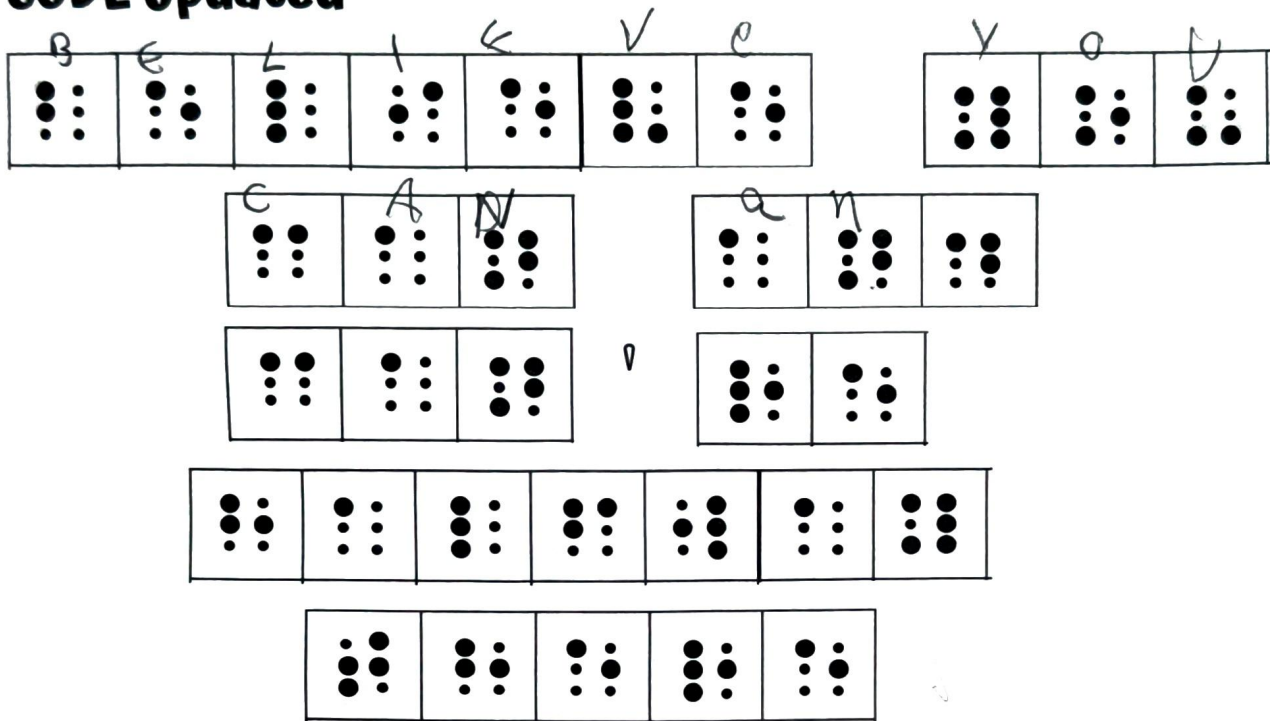
BRAILLE-DECODING

Example:



CODER

CODE Updated



DECODE THE MESSAGE

.....

.....

.....

SUDOKU

A GAME FOR MATHEMATICIANS

Fill out the blocks so that the numbers one to nine will only appear once in each row, column and 9x9 grid.

		9	7	3		5	2	6
		5		2		8		
6		8					4	7
					9		6	2
	4		6		3		8	
8	9		5					
2	6					1		8
		7		1		6		
9	5	1		6	4	2		